

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	30-07-2026	Duration:	60 minutes

Code of Conduct

I R. Pranathi (192525076) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

- 1. Study and Analyze the Scenario**
 - Carefully understand the given problem statement.
 - Identify the business domain, stakeholders, existing challenges, and system objectives.
- 2. Identify Functional and Non-Functional Requirements**
 - List all functional requirements required by the proposed system.
 - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
- 3. Identify Actors and Use Cases**
 - Identify all primary and secondary actors interacting with the system.
 - List the major use cases associated with each actor.
 - Draw a Use Case Diagram representing the interactions between actors and the system.
- 4. Prepare the Software Requirement Specification (SRS)**

Develop an SRS document containing:

 - Introduction
 - Problem Description

- Scope of the System
- Functional Requirements
- Non-Functional Requirements
- Use Case Descriptions
- Data Requirements
- External Interface Requirements
- System Features

5. Identify Assumptions and Constraints

- List the assumptions considered while developing the system.
- Identify technical, operational, economic, legal, and time-related constraints affecting the project.

6. Validate Requirement Completeness and Consistency

- Verify whether all stakeholder requirements have been addressed.
- Check for ambiguity, redundancy, inconsistency, and missing requirements.
- Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modeling	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification	SRS is well-structured, complete, professionally	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or

(SRS) Documentation	organized, and includes all required sections.			lacks essential components.
Assumptions, Constraints, and Requirement Validation	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis	/5
Functional and Non-Functional Requirement Identification	/5
Actor Identification and Use Case Modeling	/5
Software Requirement Specification (SRS) Documentation	/5
Assumptions, Constraints, and Requirement Validation	/5
TOTAL	/5

1. Introduction

The Smart Construction Project Management System is a web and mobile-based software solution developed to simplify and automate construction project management activities. The system allows project managers, engineers, contractors, clients, and workers to collaborate through a centralized platform. It provides real-time project monitoring, task management, resource allocation, equipment tracking, budget monitoring, document management, and communication. The user interface of the system is designed using **Figma**, enabling an interactive prototype before software development begins.

2. Problem Description

Traditional construction project management relies on spreadsheets, paperwork, phone calls, and manual reporting. These methods often lead to communication delays, scheduling conflicts, inaccurate resource planning, budget overruns, and project delays.

The proposed Smart Construction Project Management System addresses these problems by providing a centralized digital platform where all stakeholders can access project information in real time. It improves collaboration, reduces manual work, increases transparency, and helps complete projects on schedule.

3. Business Domain

The project belongs to the **Construction Industry** under the domain of **Project Management and Digital Construction Management**.

4. Existing Challenges

Construction companies experience several operational challenges, including:

- Manual project scheduling
- Poor communication between stakeholders
- Difficulty tracking project progress
- Delays in material delivery
- Budget overruns
- Inefficient workforce management
- Equipment maintenance issues
- Paper-based documentation
- Lack of real-time reporting
- Difficulty monitoring multiple construction sites simultaneously

5. System Objectives

The objectives of the proposed system are:

- To digitize construction project management.
- To monitor project progress in real time.
- To improve communication among stakeholders.
- To optimize workforce and equipment utilization.
- To reduce project delays.
- To improve budget tracking.
- To maintain project documentation securely.
- To generate analytical reports and dashboards.
- To support informed decision-making.
- To provide an intuitive user interface designed using **Figma**.

6. Scope of the System

The Smart Construction Project Management System manages construction projects from planning to completion. It covers project scheduling, task allocation, resource management, equipment tracking, financial monitoring, document storage, reporting, notifications, and stakeholder communication.

The system prototype and user interface are created using **Figma**, which helps visualize workflows, screens, navigation, and user interactions before implementation.

7. Stakeholders

The primary stakeholders are:

- Project Manager
- Site Engineer
- Contractor
- Client
- Workers
- Suppliers
- Finance Department
- System Administrator

8. Functional Requirements

- The system shall allow users to register and log in securely.
- The administrator shall manage user accounts and assign roles.
- The project manager shall create, edit, and manage construction projects.
- The system shall maintain project schedules.
- The project manager shall assign tasks to engineers and workers.
- The engineer shall update task progress.
- The system shall monitor construction progress.
- The system shall display project dashboards.
- The system shall manage project budgets.
- The finance department shall record project expenses.
- The system shall track construction materials.
- The system shall monitor equipment availability.
- The supplier shall update material delivery status.
- The client shall monitor project progress.
- The system shall store project documents.
- The system shall generate project reports.
- The system shall send notifications for pending tasks.
- The system shall maintain attendance records.
- The system shall generate project completion reports.
- The system shall allow project searching and filtering.

9. Non-Functional Requirements

Performance

The system shall load dashboard pages within three seconds.

The system shall support multiple users simultaneously.

Security

All passwords shall be encrypted.

Role-based authentication shall be implemented.

Only authorized users shall access confidential information.

Reliability

The system shall maintain data consistency.

Automatic backup shall be performed daily.

Availability

The system shall be available 24×7.

Scalability

The system shall support multiple construction projects simultaneously.
New users and projects shall be added without affecting performance.

Maintainability

The software architecture shall be modular.
Future updates shall be easy to implement.

Usability

The user interface shall be simple and user-friendly.
The interface prototype shall be designed using **Figma**.

10. Actors

Primary Actors

- Administrator
- Project Manager
- Site Engineer
- Contractor
- Client

Secondary Actors

- Worker
- Supplier
- Finance Officer

11. Major Use Cases

Administrator

- Login
- Manage Users
- Manage Roles
- View Reports

Project Manager

- Create Project
- Assign Tasks
- Monitor Progress
- Manage Budget
- Generate Reports

Site Engineer

- Update Work Progress
- Upload Site Images
- Report Issues

Contractor

- Manage Workforce
- Update Task Status
- Track Equipment

Client

- View Dashboard
- View Reports
- Track Progress

Supplier

- Update Material Delivery

Finance Officer

- Record Expenses

12. Data Requirements

The system stores:

- User Information
- Project Details
- Task Details
- Employee Information
- Equipment Records
- Material Inventory
- Budget Information
- Expense Records
- Attendance Records
- Project Reports
- Uploaded Documents
- Notifications

13. External Interface Requirements

User Interface

The complete application interface is designed using **Figma**, including:

- Login Page
- Dashboard
- Project Management
- Task Management
- Resource Management
- Equipment Tracking
- Budget Dashboard
- Reports
- Notifications
- User Profile

Hardware Interface

Desktop Computer

Laptop

Tablet

Mobile Device

Software Interface

Figma (UI Design)

MySQL Database

Web Browser

Cloud Storage

14. System Features

The proposed system includes:

- Secure Authentication
- Dashboard Analytics
- Project Monitoring
- Task Assignment
- Resource Allocation
- Equipment Management
- Material Tracking
- Budget Monitoring
- Document Management

- Notification System
- Report Generation
- Role-Based Access Control

15. Assumptions

All users have internet access.

Construction companies adopt digital workflows.

Users possess valid login credentials.

Construction data is entered accurately.

Project managers regularly update project information.

Figma is used to design the complete application prototype before development.

16. Constraints

Technical Constraints

Internet connectivity is required.

The system depends on cloud infrastructure.

Operational Constraints

Users require basic computer knowledge.

Training may be necessary.

Economic Constraints

Software development and maintenance involve implementation costs.

Legal Constraints

The system must comply with organizational privacy and security policies.

Time Constraints

The project must be completed within the academic semester.

17. Requirement Validation

- The proposed requirements satisfy the needs of all stakeholders.
- All functional requirements support the project objectives.
- Non-functional requirements improve system quality.
- Actors and use cases cover all major business processes.
- No duplicate or conflicting requirements are identified.
- The system architecture supports future scalability.
- The **Figma prototype** accurately represents the navigation, user interfaces, workflows, and module interactions, reducing design ambiguity before software development.

FIGMA



SMART CONSTRUCTION PROJECT MANAGEMENT SYSTEM

Manage projects, resources,
materials, and progress
from one centralized platform.



Welcome Back

Sign in to access your
construction management dashboard.

LOGIN

☒ Remember Me

[Forgot Password?](#)

SCMS

Smart Construction Management System

Dashboard

Projects

Employees

Resources

Materials

Progress

Reports

Settings

Logout

Dashboard

Overview of projects, activity, and system events

Search projects, employees...

Admin

Welcome Back, Admin

Monitor construction projects, resources, budgets, and progress in real time.

Total Projects

24

Across all sites

Active Projects

12

In progress

Completed

08

Handover ready

Delayed

04

Needs attention

Project Status Overview

5 active projects with completion percentages

Highway Bridge Expansion

45%

Downtown Office Tower

68%

School Campus Build

85%

Residential Complex Phase 2

100%

Metro Station Renovation

30%

Recent Activity

Pranathi uploaded Safety Report

2 hours ago

Suresh approved material request

4 hours ago

Geetha completed site inspection

Yesterday

Hema logged equipment maintenance

Yesterday

New material delivery received

2 days ago

SCMS

Smart Construction Management System

Dashboard

Projects

Employees

Resources

Materials

Progress

Reports

Settings

Logout

Projects Administration

Search projects, employees...

Admin

TOTAL PROJECTS

24

ACTIVE PROJECTS

12

COMPLETED

08

DELAYED

04

Status: All Projects

Sort By: Newest

+ Add New Project

Project Name	Location	Status	Progress	Budget	Start Date	End Date
Highway Bridge Expansion	District 4 Outer Ring	Active	45%	\$12.4M	12 Jan 2025	30 Nov 2026
Downtown Office Tower	Plaza Central	Active	68%	\$45.0M	01 Aug 2024	15 Dec 2026
Residential Complex Phase 2	Westside Meadows	Completed	100%	\$18.2M	15 May 2023	10 Jan 2025
Metro Station Renovation	Underground-Line 2	Delayed	30%	\$8.9M	10 Oct 2024	20 May 2026
Industrial Warehouse	Logistics Park East	Active	15%	\$6.5M	01 Mar 2025	01 Feb 2026
School Campus Build	Community Zone 9	Active	85%	\$14.0M	15 Feb 2024	15 Aug 2025
Commercial Plaza Parking	Sunset Boulevard	Completed	100%	\$3.1M	01 Jun 2024	30 Dec 2024

SCMS

Smart Construction Management System

Dashboard

Projects

Employees

Resources

Materials

Progress

Reports

Settings

Logout

Employee Directory

TOTAL EMPLOYEES

186

ON-SITE ACTIVE

124

ON-LEAVE

18

NEW HIRES (MONTH)

12

Construction Workforce Staffing

Name	Role	Assigned Project	Status	Contact Email
Pranathi	Site Architect	Downtown Office Tower	Active	pranathi@scms.com
Sonu	Chief Foreman	Highway Bridge Expansion	Active	sonu@scms.com
Suresh	Structural Engineer	School Campus Build	Active	suresh@scms.com
Geetha	Safety Inspector	Metro Station Renovation	On Leave	geetha@scms.com
Hema	Lead Mason	Residential Complex Phase 2	Active	hema@scms.com
Abhighna	BIM Specialist	Industrial Warehouse	Active	abhighna@scms.com
Roopa	Heavy Machine Operator	Highway Bridge Expansion	Active	roopa@scms.com

SCMS

Smart Construction Management System

Dashboard

Projects

Employees

Resources

Materials

Progress

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Heavy Equipment & Material Inventory

Heavy Machinery & Fleet Status

Asset/Equipment	Assigned Project	Availability
Tower Crane TC-01	Downtown Office Tower	In Use
Crawler Excavator EX-04	Highway Bridge Expansion	In Use
Concrete Mixer CM-08	School Campus Build	Available
Bulldozer BD-02	Residential Complex Phas...	Maintenance
Soil Compactor SC-05	Metro Station Renovation	Available

Raw Materials Stock Ledger

Portland Cement (Grade 53)

1,200 Bags

Structural Steel Bars (Reinforced)

450 Tons

Red Clay Bricks

2,500 Units

▲ Low Stock - Reorder Raised

River Sand (Fine Aggregate)

800 m³

Cured Structural Timber

120 Units

▲ Critical Stock - Urgent Reorder

SCMS

Smart Construction Management System

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Materials Inventory & Procurement

Search projects, employees...

Admin

TOTAL MATERIALS45

IN STOCK32

LOW STOCK8

OUT OF STOCK5

Category: All

Status: All Stock States

+ Add Material

Material Name	Category	Quantity	Unit	Supplier	Status	Last Ordered
Portland Cement	Binding	1200	Bags	UltraTech Cement	In Stock	12 Jan 2026
TMT Steel Bars	Structural	450	Tons	Tata Steel	In Stock	08 Jan 2026
Red Clay Bricks	Masonry	124	Units	Local Supplier	Low Stock	15 Jan 2026
River Sand	Aggregate	800	m³	Krishna Aggregates	In Stock	10 Jan 2026
Structural Timber	Woodwork	120	Units	Greenwood Supplies	Out of Stock	03 Dec 2025
Glass Panels	Finishing	340	Sheets	Saint-Gobain	In Stock	14 Jan 2026
PVC Pipes	Plumbing	560	Pieces	Finolex	Low Stock	11 Jan 2026

SCMS

Smart Construction Management System

Dashboard

Projects

Employees

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Logout

Project Progress Analytics & Documentation

Search projects, employees...

Admin

Planned vs. Actual Progress (Top 5 Projects)

Downtown Office Tower

75% Planned

68% Actual

School Campus Build

80% Planned

85% Actual

Residential Complex Phase 2

100% Planned

100% Actual

Highway Bridge Expansion

55% Planned

45% Actual

Metro Station Renovation

50% Planned

30% Actual

Recent Generated Reports

Weekly Progress Report

Generated: May 12, 2025

PDF • 2.4 MB

Budget Summary Ledger

Generated: May 10, 2025

PDF • 4.1 MB

Safety & Compliance Log

Generated: May 08, 2025

PDF • 1.2 MB

Resource Utilization Audit

Generated: May 01, 2025

PDF • 3.8 MB

 **SCMS**
Smart Construction Management System

Dashboard

Projects

Employees

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Progress

Reports

Settings

Logout

TOTAL PROJECTS
24

ACTIVE PROJECTS
12

COMPLETED
8

DELAYED
4



Confirm Logout

Are you sure you want to log out of SCMS? You will need to re-verify your administration credentials to gain system access.

Cancel

Logout